CHAPTER 21

The Nuclear Fuel Cycle

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The nuclear fuel cycle comprises the handling of all fissile and fertile material necessary for nuclear power production and of the radioactive products formed in this process (Fig. 21.1). The fuel cycle is suitably divided into a front end and a back end part, where the nuclear power station is the dividing line. The front end comprises uranium exploration, mining, and refining (§14.2.4), isotope enrichment (§3.8), and fuel element fabrication (§21.1). Reactor operation involves fuel behaviour during operation, canning corrosion etc., while the back end involves reprocessing and radioactive waste ("radwaste") handling. Health and environmental aspects are important in all these steps, but being of a more general nature (see Ch. 15 and 22), they are not considered as "steps" in the nuclear fuel cycle.

The nuclear fuel used in almost all commercial reactors is based on uranium in the form of UO_2, either enriched so that the ^{235}U content has been increased to a few percent, or – less commonly – with the natural 0.7% abundance of the fissile isotope ^{235}U. Some power reactors also use fuel containing depleted uranium ($\sim$0.3% ^{235}U) in which plutonium is bred and/or Pu mixed with ^{238}U as a replacement for ^{235}U ("mixed oxide fuel"). ^{232}Th, in which fissile ^{233}U is bred, has also been used in a few cases.

The fuels for the next generation of reactor systems are typically either, oxides, nitrides, carbides, inert matrix or metallic. All these fuel types have distinct advantages and disadvantages raging from ease of manufacturing to direct safety issues by themselves or in combination with the selected coolants in the reactor. The coolants often discussed for future fast reactors are: molten sodium, molten lead or a gas such as helium or carbon dioxide. They also have their distinct pros and cons. Currently the majority of fast reactor hours have been run with liquid sodium as coolant even if its interactions with water is one of the major draw backs. In e.g. Russia a considerable experience with liquid lead, or liquid lead–bismuth, as coolant exist, but it has probably caused at least one accident in a nuclear submarine. In this case violent reactions with water are avoided but instead significant erosion and corrosion of e.g. coolant pumps exist.

Whether based on uranium, thorium, or plutonium, a fuel must be capable of resisting temperatures considerably above 1000°C without physical or chemical deterioration due to heat or to radiation. Nitride based fuels are of interest for future fast reactors due to their good heat conductivity. However, due to the internal N_2-pressure they need another canning material than zircaloy. Metallic fuels have the high heat conduction necessary to minimize temperature gradients. Uranium melts at 1130°C and plutonium at 640°C. However, metallic uranium has three and plutonium six allotropic forms between room temperature and their melting points. As a consequence, either the separate or combined effects of the radiation field, the high pressure during operation,

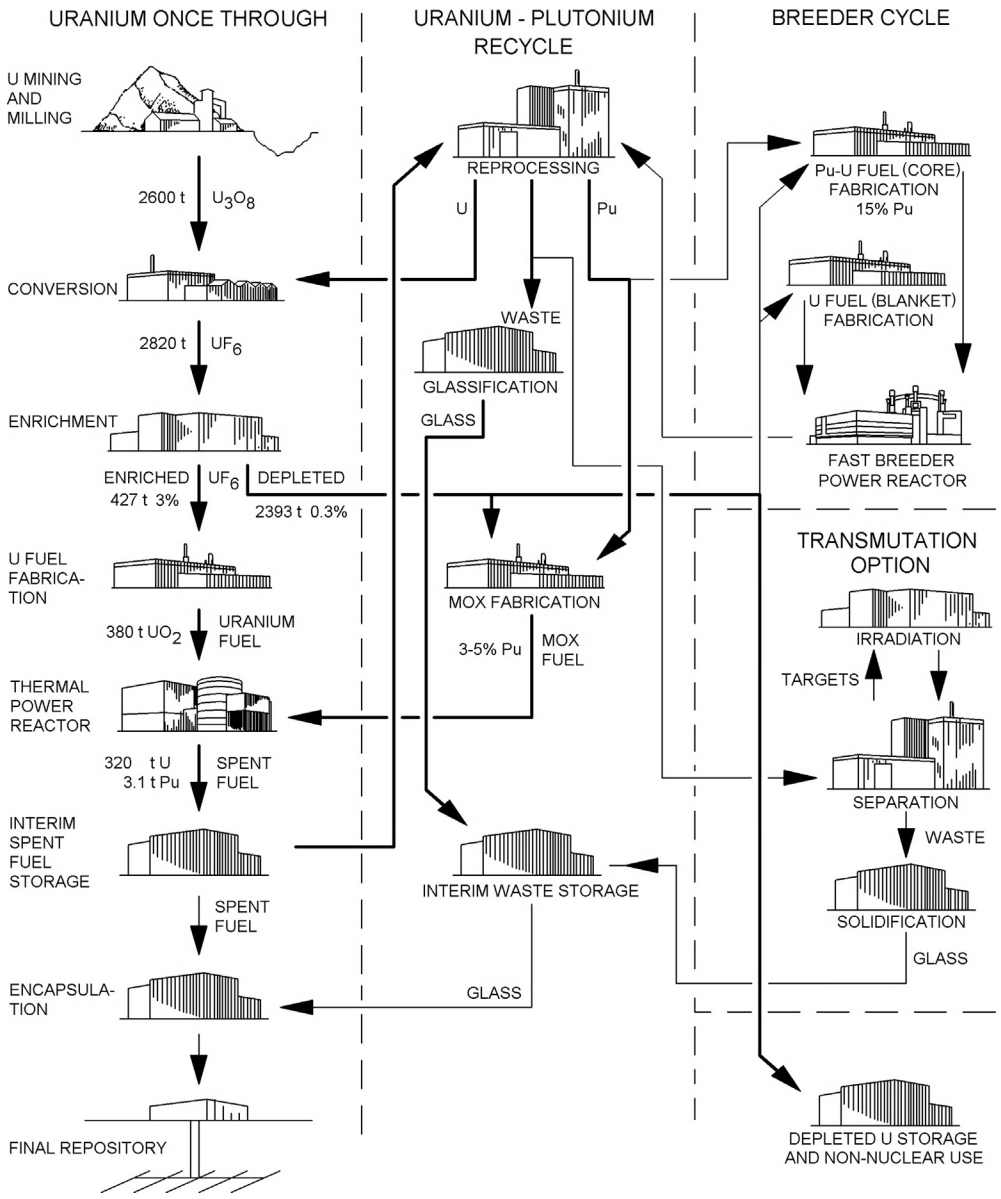


Figure 21.1 Fuel cycle alternatives. Annual flows of materials in a 10 GW_e LWR UOT program are indicated.

and the high temperature can cause recrystallization into different allotropic forms with significantly different volume. Volume changes within the fuel element during operation cause mechanical deformations, reduce the mechanical strength, and increase the problem of corrosion even if the elements are clad in another corrosion resistant metal.

With the exception of some older gas cooled reactors, power reactors use ceramic pellets of UO_2, PuO_2, and ThO_2, or a mixture of these oxides, as fuels. UC has also been tested in some reactors. The size of the cylindrical pellets is $\sim 1 \times 1$ cm (diameter $\times$ height). To improve the possible burnup and to increase the temperature margin in the pin centers, the pellets are now often thinner in diameter, e.g. 0.8 cm. Fuel rods consisting of ceramic fuel pellets stacked in metallic tubes of zircaloy or stainless steel are quite temperature resistant, do not have the phase transformations of the metals, and have greater resistance to radiation effects. Unfortunately, the heat conduction is not as good as in the metallic fuel elements, and as a result rather high temperature gradients (up to $300°C\ mm^{-1}$) often exist in ceramic UO_2 elements.

21.1. PRODUCTION OF FUEL ELEMENTS

The normal raw material for production of UO_2 based fuels is enriched uranium in the form of UF_6, which is delivered in special containers. By heating the container to $\sim 100°C$ it is possible to transfer the hexafluoride as gas to the conversion plant, where UO_2 powder is produced, see Fig. 21.2. Several possible reactions can be used, e.g. hydrolysis of UF_6 by dissolution in water

$$UF_6(g) + 2H_2O \rightarrow UO_2F_2(aq) + 4HF(aq) \quad (21.1)$$

followed by precipitation with ammonia

$$2UO_2F_2(aq) + 6NH_4OH(aq) \rightarrow (NH_4)_2U_2O_7(s) + 4NH_4F(aq) + 3H_2O \quad (21.2)$$

after filtration, washing and drying the diuranate is converted to UO_2 by reduction with hydrogen at 820°C.

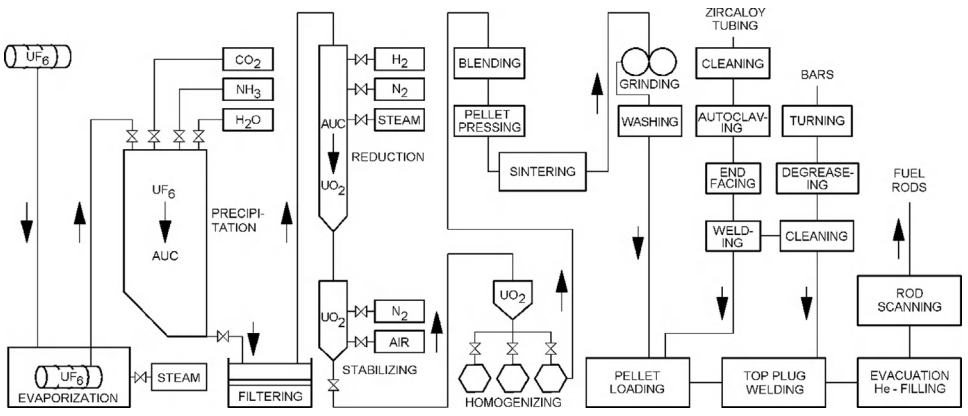


Figure 21.2 Production of UO_2-fuel rods.